INCH-POUND

MIL-G-53133/10
30 August 1993

MILITARY SPECIFICATION SHEET

GENERATOR SETS, SKID MOUNTED, TACTICAL, QUIET,

DIESEL ENGINE DRIVEN, ALTERNATING CURRENT

60 KILOWATTS, 400 HERTZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-G-53133.

1. SCOPE

1.1 Scope. This specification sheet covers the detail requirements for tactical generator sets, size 60 kW, mode II (400 Hz), model MEP-816A, NSN 6115-01-274-7395.

2. APPLICABLE DOCUMENTS

2.1 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

TA 30554-88-816	- Generator Set, Tactical, Quiet, 60 kW, 400 Hz.
30554-88-23004	- Instructions, Preservation and Packing, 60 kW.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-FGM, Fort Belvoir, VA 22060-5606.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-G-53133 as specified herein.

AMSC N/A FSC 6115

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3.2 Description. The set shall be in accordance with TA 30554-88-816 and as specified herein.

3.3 Reliability. The minimum acceptable mean time between failure (MTBF) for the set shall be 361 hours with a 80 percent lower confidence limit.

3.4 Size and weight. The maximum operating set dimensions with control panel doors closed and the total operational weight shall not exceed the following:

Parameter	Dimension
Length	2192 mm (86.3 inches)
Width	897 mm (35.3 inches)
Height	1478 mm (58.2 inches)
Weight	1884 kg (4153 pounds)

3.5 Fuel consumption. Fuel consumption, at all loads up to and including rated load, shall not be more than 22.74 ℓ/h (6.0 gallons per hour [GPH]) when the set is operating on referee grade diesel fuel conforming to MIL-F-46162.

3.6 Parallel operation. With their governor and voltage regulator paralleling circuits interconnected, two sets of the same mode and size operated in parallel shall divide load in accordance with the following as system load (at rated power factor) is varied between zero and 100 percent (and vice versa) of the combined rating of the connected sets:

a. Real power division. At no time shall difference between the average individual kilowatt outputs of the set, when supplying the system load, be greater than 10 percent of the kilowatt rating of one set.

b. Power exchange. At all constant system loads up to the combined rating of the sets in parallel, power exchange between the sets shall not exceed 10 percent of the kilowatt rating on one set. Power exchange is the difference between the maximum and minimum power output delivered by one set for constant system load conditions, as determined from oscillographic measurements.

c. Reactive power division. At no time shall the difference in average reactive kilovolt-ampere (kVA) supplied by any two set differ by more than 10 percent of the reactive kVA rating on one set.

For the above requirements, the initial system load shall be equally divided between the sets, both as to activate and reactivate components; thereafter, there shall be no adjustments to governors, voltage regulators, or any other components as system load is changed.

3.7 Frequency regulation. The system shall provide isochronous operation, i.e., for every load change (including rated load) the frequency regulation shall not exceed 0.25 percent of rated frequency.

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3.8 Frequency short-term stability (30 seconds). At every constant load from no load to rated load, the system shall maintain frequency within a bandwidth equal to 0.5 percent of rated frequency. The system shall not permit repetitive frequency variations, commonly called "hunting", even though within the allowable 0.5 percent band.

3.9 Frequency long-term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant voltage, and any constant load from no load to rated load, the system shall maintain frequency within a bandwidth equal to 1 percent of rated frequency in a 4-hour operation period.

3.10 Frequency drift. With the set operating at constant load and voltage, a change in ambient temperature of up to 33 C° (60 F°) in an 8-hour period shall not cause the frequency to change by more than 0.5 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.11 Frequency transient performance. Following a sudden increase or decrease in load up to and including rated load, the governing system shall reestablish stable engine operating conditions within 1 second. The maximum transient frequency change above and below the new steady-state frequency shall be not more than 1.5 percent of rated frequency.

3.12 Frequency adjustment. By means of the frequency adjust device, it shall be possible to adjust the frequency, (when the set is operating at any constant load) between 390 and 420 Hz but not below 370 Hz nor above 430 Hz. It shall not be possible to adjust frequency to a value which will actuate the overspeed protective device.

3.13 Voltage waveform. The deviation factor for the line-to-line and line-to-neutral voltage waveform for each voltage connection shall be not more than 5 percent, nor shall any single voltage harmonic be more than 2 percent. In addition, there shall be no evident discontinuities, spikes, or notches in the waveform when viewed on an oscilloscope having a bandwidth of DC to 1.5 MHz and usable viewing screen of 8 x 10 cm. The oscilloscope gain shall be adjusted such that one cycle of voltage wave covers approximately the entire viewing screen.

3.14 Voltage unbalance. With the set operating at rated voltage and frequency and no load, maximum difference between line-to-line voltages shall be not more than 5 percent of rated voltage for the 240/416-volt connection under the conditions of a single phase, line-to-line unity pf-load of 25 percent rated current and no other load on the generator.

3.15 Phase balance voltage. The maximum difference in the three line-to-neutral voltages shall be not more than 1 percent of the rated line-to-neutral voltage. The maximum difference between the voltage of the 120-volt windings of any phase shall be not more than 1 volt.

3.16 Voltage modulation. The voltage modulation of the output voltage for any load up to and including rated load shall not exceed 1 percent. This requirement shall apply to all line-to-line and line-to-neutral voltages.

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3.17 Voltage regulation. The voltage regulation from no load to rated load and from rated load to no load shall be not more than 1 percent of the rated voltage for all voltage connections throughout the voltage ranges specified in 3.22.

3.18 Voltage short-term stability (30 seconds). At every constant load from no load to rated load, the voltage at the set terminals shall remain within a bandwidth equal to 1 percent of rated voltage.

3.19 Voltage long-term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant frequency and any constant load from no load to rated load, the voltage shall remain within a bandwidth of 2 percent of rated voltage in a 4-hour operating period.

3.20 Voltage drift. With the set operating at constant load and frequency, a change in ambient temperature up to 33 C° (60 F°) in an 8-hour period shall not cause the voltage to change by more than 1 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.21 Voltage transiet performance. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows (reactive droop compensation of the voltage regulator being inoperative):

a. With the set initially operating at no load, rated voltage, and rated frequency, the terminal voltage of the set shall not drop to less than 75 percent of no load voltage when a balanced 3-phase, 0.4 pf (or less) lagging, static load having an impedance of 0.5 per unit is suddenly applied to the output terminals of the set. When connected to the specified load, the output voltage shall recover to a minimum of 95 percent of rated voltage within 0.7 seconds and shall stabilize at or above this voltage. The above specified voltage dip shall not be exceeded when a fully-loaded induction motor of the above specified impedance is used in place of a static load, and no reactions shall be set up to prevent full acceleration of the motor with rated torque applied to its shaft.

b. When the set is initially operating at rated frequency, rated voltage, and following any sudden change in load from no load to rated load, the instantaneous voltage shall not drop to less than 88 percent of rated voltage and shall reach stable conditions within 0.5 seconds; no overshoot or undershoot of the final voltage shall not exceed the initial voltage transient in amplitude. The above requirements shall also apply when the load is suddenly changed from rated load to no load, except that the initial voltage transient shall not exceed 112 percent of rated voltage.

c. The generator set shall be capable of across-the-line starting a motor rated at 1 horsepower per kW of set kW rating. The starting current rating of the motor shall be NEMA Code F and the motor shall be loaded with a flywheel having an inertia equal to that of the motor rotor.

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3.22 Voltage adjustment range. The minimum voltage adjustment range shall be:

Voltage Connection	Range
120/208 V	197 to 229 volts
240/416 V	395 to 458 volts

This voltage adjustment range shall apply for all load between no load and rated kW, rated power factor at rated frequency under all environmental conditions. It shall not be possible to adjust the voltage of a value which activates the overvoltage protective device.

3.23 Endurance. The set shall operate for 3,000 hours without critical failure at all loads, continuous and intermittent, up to and including rated load under all of the individual or combination of conditions specified in 3.9.1 of the general specification.

3.24 Exciter field current. With the generator temperature stabilized at normal ambient conditions, the exciter field current shall not exceed 1.78 amperes for rated load at 400 Hz operation.

4. PACKAGING

4.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with 30554-88-23004.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Army - ME

Project 6115-0274

Review activities:
Navy - MC
DLA - GS

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